How Does Emissions-Charging Influence House Prices? Evidence From London's ULEZ

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Abstract

In 2019, the Ultra Low Emissions Zone (ULEZ) was implemented in London: a daily charge on all non-compliant cars driven in a specified zone. This zone was expanded in 2021 and again in 2023. I construct a novel dataset on UK house sales, and use a staggered difference-in-differences design to find that the implementation of the ULEZ led house prices in these expansion zones to fall on average by roughly 3-4% over the following year. This corresponds to a windfall loss to the average London homeowner of roughly £25,000, or half the median annual salary in the region. This result is robust to considerable weakening of the parallel trends assumption. It is also robust to different methods of dealing with spatial spillovers, and I quantify the spillover effects to find that the ULEZ leads prices of houses just outside the region to fall by roughly 2% over a similar time-frame. Beyond two years after implementation, the main negative effect begins to recover, which I justify as a product of asymmetric information transmission.

JEL Classification: C23, Q53, Q58, R21, R31

Keywords: emissions charges, house prices, difference-in-differences, environmental policy, spatial spillovers

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The online appendix and other resources can be found at https://github.com/jacobmcl7/ULEZ_house_prices_WMESP

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1 Introduction

In 2018, the European city suffering the worst costs from air pollution was London – they were 79.5% higher than second-placed Bucharest (CE Delft, 2020). To address this, the Mayor of London established the Ultra Low Emissions Zone (ULEZ), within which drivers of non-compliant vehicles are charged £12.50 per day. The ULEZ is one of the largest and strictest emissions-charging zones worldwide, making a comprehensive understanding of its overall impact crucial. Importantly, this must extend further than its intended effects on air pollution, to include both positive (e.g. reduced traffic and noise) and negative (e.g. increased transport costs, firm relocation) unintended effects.

The best way to ascertain the overall effect of the policy on affected residents is to examine its effect on the local quality of life, proxied for by local house prices. This connection is grounded in the hedonic pricing model, where house prices are modelled as the sum of the value of various internal and external components, including air quality, congestion, and transportation costs. Changes in these individual valuations change house prices accordingly: these aggregate every possible dimension of the ULEZ effect together. This paper therefore seeks to answer the following question: how did the ULEZ affect house prices in London?

By creating a novel dataset through combining house sale data from HM Land Registry with geographic information system (GIS) software, I find that the ULEZ led house prices to fall by roughly 3-4%. This result suggests the loss of freedoms due to travel restrictions far outweigh any gains from, for example, improved air quality and health. The effect follows a U-shaped profile over time – relative prices fall sharply, plateau out at around 4%, then begin to recover, which is rationalizable as a product of asymmetric information transmission. I also find that areas just outside the ULEZ experienced spillover effects of around a 2% drop in prices.

The paper proceeds as follows. Section II explains the charge and its expansion process, and section III reviews the literature on the broad spectrum of effects of emissions charges in London and elsewhere. Section IV outlines the data collection and processing, followed by a methodological outline in section V. Section VI presents my results, discussed in section VII, and section VIII concludes.

2 Background

The ULEZ, first announced in March 2015, was implemented in April 2019 in a 20km^2 region in Central London, with the explicit aim of improving air quality (Transport for London, 2025).

The policy imposes a daily £12.50 charge on all vehicles driven in the zone that do not adhere to certain environmental standards. The ULEZ expanded in October 2021 to the North and South circular roads, as announced in June 2018. It then expanded again in August 2023, as announced in November 2022, to cover almost all 1,572km^2 of Greater London, where it remains today. I illustrate these expansion zones in Figure 1.

The original 2019 ULEZ differed in several ways to the zones it later expanded into. Firstly, until the 2021 expansion, residents of the 2019 zone were exempt from the charge. Further, the 2019 zone had already been subject to some emissions-related driving restrictions – the Toxicity Charge (operating 2017-2019) and the Congestion Charge (operating 2003-present). I therefore disregard the 2019 zone from my analysis, as residents had a fundamentally different experience of the ULEZ to those in the later expansion zones. Aside from a separate Low Emissions Zone aimed at commercial vehicles, the 2021 and 2023 expansion zones had never experienced emissions-charging, and so their experience resembles the conventional introduction of a 'traditional' (in the sense that residents are charged) emissions charge.

Since 2019, the charge has remained fixed at £12.50, as have both the targeted vehicles and the environmental standards. As such, ignoring spillover effects, the 'treatment' of London's houses to the ULEZ has been homogeneous, although staggered, which I leverage in my analysis.

3 Literature Review

This paper contributes to the emerging intersection between environmental and urban economics, by building on the literature surrounding the effects of emissions-charging policies. In particular, the direct effects of such charges are well-evaluated. The gradual implementation of broadly similar LEZs in German cities is naturally conducive to event-study designs, which have identified their positive effect on air quality (Wolff, 2014) and health outcomes (Pestel and Wozny, 2021; Margaryan, 2021). Similar findings have emerged from LEZ-type policies in other European cities (Simeonova et al, 2019, Panteliadis et al, 2014, Bernardo et al, 2021). In London, the effects of the Congestion Charge (LCC) and the Low Emission Zone (LEZ) have been more ambiguous, though potentially stemming from simply their limited scope. Zhai and Wolff (2021) and Ellison et al (2013) fail to find much evidence supporting the effect the LEZ had on air quality, and Green et al (2020) share this conclusion for the LCC. Ma et al (2021) extend this ambiguity to the 2019 implementation of the ULEZ, but London City Hall (2024) suggest that the expansions were considerably more successful. In general, large emissions charges achieve their main objective.

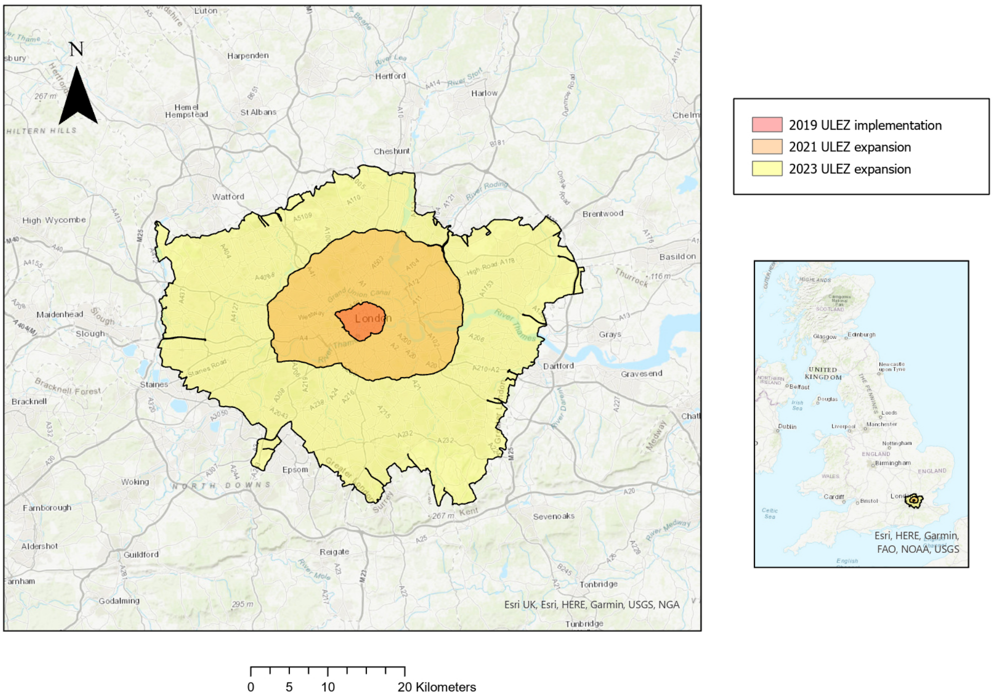


Figure 1: The geography of the ULEZ, as it was implemented and then expanded

There is less consensus on their indirect effects. An important one is road congestion, and Moral-Carcedo (2024) finds that Madrid's LEZ led congestion to fall inside the zone, but rise outside it. Studies in Germany find no effect on traffic volumes inside or outside the zone (Pestel and Wozny, 2021; Wolff, 2014). However, Herzog (2024) constructs an 'exposure index' to London's LCC, and uses it in a difference-in-differences framework to find it reduced traffic both in the charge zone and on suburban roads heading in the zone's direction. He then builds this into a structural model which suggests positive distributional effects of the LCC, since the areas benefitting most from the traffic reductions house predominantly low-skilled jobs. This feature of London's structure may imply similarly positive secondary effects for the ULEZ.

House prices are often understood through hedonic pricing frameworks, established in Rosen (1974), within which they are determined through the sum of the values of distinct internal and external components, including air quality, congestion and local transportation costs. However, while research has established the connection between these individual components and house prices (particularly air quality – e.g. Chay and Greenstone, 2005), very few studies have convincingly collected them together to assess the overall impact of emissions-charging. Jabbar et al (2024) finds a positive effect of the 2019 introduction of the ULEZ on house prices, but uses a sharp RDD with the cutoff at the zone's border, therefore ignoring spatial spillovers. The zone's peculiarities mean this effect is also not reflective of more conventional emissions-charges. In Milan, Percoco (2014) finds the Ecopass emissions charge reduced house prices in the region by roughly 1.5%, but fails to account for spillovers, anticipation, and the dynamics of the result. Gruhl et al (2022) use a stacked difference-in-difference design to find a positive effect of Germany's many LEZs on apartment rents, as well as other housing types, but again their results change when adjusting for spatial spillovers. An alternative evaluation of Germany's LEZs on overall welfare comes from Sarmiento et al (2023), who conversely find that they led residents' subjective wellbeing to fall. Convincing literature on the overall effects of an emissions charge is sparse - an important and very policy-relevant gap this paper aims to fill.

This paper contributes to the discourse on the effects of emissions-charges, by using a novel dataset to provide a holistic overview of one of the largest and strictest - London's ULEZ. Given the increasing policy-relevance of emissions-charging, this analysis matters both because of the scale of the ULEZ, and because existing literature generally ignores unintended secondary effects. This paper also makes a methodological contribution, applying recent theoretical research to handle both spatial spillovers and parallel trends more comprehensively.

4 Data

In my analysis, I use the Price Paid Data from HM Land Registry, from which I take the sale price, address and characteristics of all houses sold in England and Wales from January 2015 to November 2024. I then use ArcGIS to relate these sales to the ULEZ expansion, by geocoding each sale address and calculating its shortest planar Euclidean distance to each ULEZ border, then signing the distances according to whether the house is within the border in question (negative distances signify houses within the border). Full details of the data collection and processing, are in Appendix 1. Keeping houses within 20km of the current ULEZ border leaves me with 1,308,297 observations, whose numeric characteristics are summarised in Table 1. The data provides relatively few house characteristics, only giving indicators for the house type, lease type, and whether it was newly built at the point of sale. However, while this could imperil my identifying assumptions, I harness new econometric research to illustrate the robustness of my results to violations of these assumptions. My lack of controls isn't a big problem.

My unit of analysis is the house, but I consider treatment (i.e. status as within ULEZ or not) to

VARIABLES	N	mean	sd	min	max	Source
Log(sale price)	1,308,297	13.06	0.586	0	19.95	Contained in the data
Detached?	1,308,297	0.119	0.324	0	1	Contained in the data
Semidetached?	1,308,297	0.190	0.392	0	1	Contained in the data
Terraced?	1,308,297	0.269	0.443	0	1	Contained in the data
Flat?	1,308,297	0.422	0.494	0	1	Contained in the data
New?	1,308,297	0.113	0.317	0	1	Contained in the data
Leasehold?	1,308,297	0.432	0.495	0	1	Contained in the data
In 2019 ULEZ?	1,308,297	0.016	0.123	0	1	Constructed with ArcGIS
Distance from 2019 ULEZ (km)	1,308,297	17.62	12.08	-1.942	46.42	Constructed with ArcGIS
In 2021 ULEZ?	1,308,297	0.255	0.436	0	1	Constructed with ArcGIS
Distance from 2021 ULEZ (km)	1,308,297	9.427	11.64	-9.191	37.82	Constructed with ArcGIS
In 2023 ULEZ?	1,308,297	0.611	0.488	0	1	Constructed with ArcGIS
Distance from 2023 ULEZ (km)	1,308,297	-0.137	9.089	-14.76	20.00	Constructed with ArcGIS

Table 1: Simple summary statistics for the numeric variables in the data

be assigned at the postcode-sector level[1]. Occasionally, postcode sectors overlap ULEZ boundaries. I split any such postcode sector in two at the border, treating them as separate. The data contains house sales from 1705 (modified) postcode sectors over 40 quarters, the default time period I consider.

The data needs further refinement for baseline analysis. Since I focus on the effect of the 2021 and 2023 expansions, I drop all observations within the original 2019 zone, but more steps are required to ensure important identifying assumptions hold. People living outside the zone pay the charge each time they drive into it, and any improvements in air quality, congestion and other relevant factors will all percolate out of the zone's border, so people living in houses near an active ULEZ will experience similar effects to those within the zone – spatial spillovers are an issue. To address this, I remove all house sales outside, but within 5km from, the 2019, 2021 and 2023 borders. These will be most susceptible to spillovers and so by removing them I remove the bulk of the relevant bias from the analysis. The idea of assuming a reasonable distance beyond which spillovers end is a common way to define treatment and control groups (e.g. Gupta et al, 2022), and I take the 5km figure from Gruhl et al (2022) who use it for the same purpose in their robustness checks. If spillovers extend past 5km from the zone, they will bias the estimated treatment effect towards zero, meaning any significant results found suggest true significance, forming a lower bound for the true effect.

[1]UK postcodes take the form AB1 2CD. A postcode sector is the collection of houses that share the AB1 2xx component of their postcode.

I must also address the representativeness of my controls. House prices in urban and rural areas evolve differently over time (Glaeser et al, 2001), and since ULEZ has been implemented in disproportionately urban areas, the control group cannot be too rural. To address this, I remove observations at least 10km outside the 2023 border. This figure aims to balance between leaving the control group too large that it becomes unrepresentative, and too small that sample sizes cause issues with precision. These series of refinements then leave me with three rings of observations, as shown in Figure 2. The size of each group, and the quarter they were first exposed to the ULEZ, are given in Table 2. These three regions form the basis of my analysis.

5 Methodology

I aim to examine the impact that the ULEZ expansions had on house prices in affected regions. To form a prior over what I expect to find in the analysis, I use kernel density smoothing to approximate the distributions of house prices in each of the two expansion regions, before and after the relevant expansion, and compare them to the same distribution in the control zone before and after the same point in time, as plotted in Figure 3. As would be expected, inflation shifts the distributions rightwards over time. There is very little difference in the 2023 comparison, but the 2021 comparison shows a larger rightward shift in the control zone than the 2021 zone, particularly around the peak of the distribution. These figures suggest a drop in house prices associated with the implementation of ULEZ. More rigorous analysis is required isolate this fall and deal with causality.

On this last point, it is crucial to first note that a naive comparison of house prices within and outside of an active ULEZ would be vulnerable to endogeneity, with selection issues coming from the fact that houses exposed to ULEZ are closer to central London, or closer to better public transport, for example. It is hard to control for, or even observe, all these relevant factors.

To more convincingly isolate a causal effect, I utilise the staggered, homogeneous expansion of the ULEZ, which lends itself to a staggered difference-in-differences (DiD) design. As a baseline, I estimate the static treatment effect in the standard two-way fixed-effects (TWFE) form, estimating the following equation by OLS:

$$y_{ipt} = \alpha_p + \delta_t + \beta ULEZ_{pt} + \rho \mathbf{X}_{it} + \varepsilon_{ipt}$$

where y_{ipt} is the log of the sale price of house i in postcode sector p, sold in quarter t, with postcode-sector and time fixed-effects α_p and δ_t, a vector of covariates $\mathbf{X}_{it}$ (given in Table 1), and $ULEZ_{pt}$ an indicator for whether the postcode sector containing house i is within the ULEZ

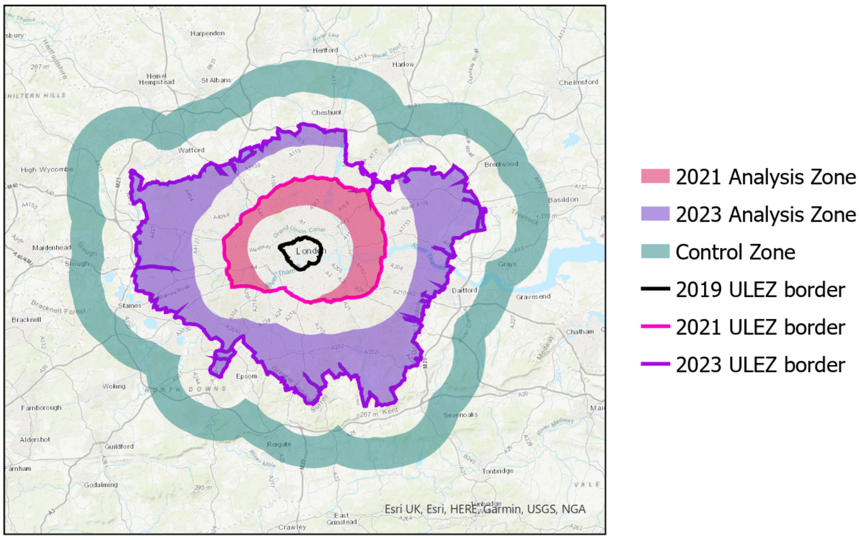


Figure 2: Geographical representation of the three groups/regions in the analysis

Zone	**N**	**First treatment**
2021 Analysis Zone	123,788	2021 Q4
2023 Analysis Zone	243,546	2023 Q3
Control Zone	114,258	-

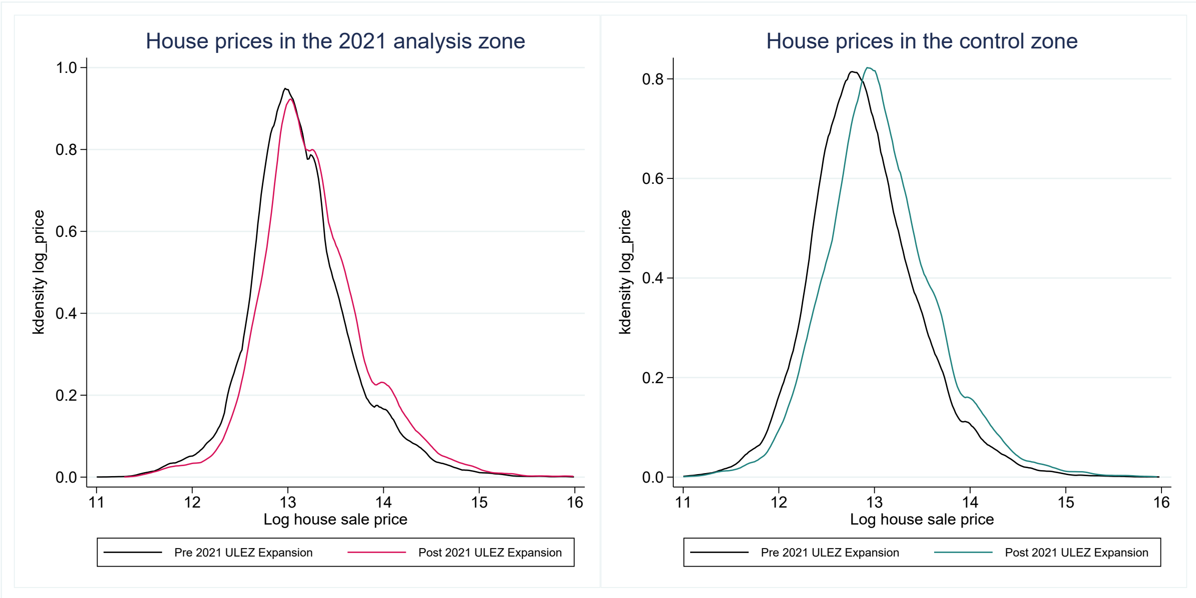
Table 2: Overview of the key characteristics of each group

in quarter t. The coefficient of interest, β, represents the 'extensive margin' overall treatment effect.

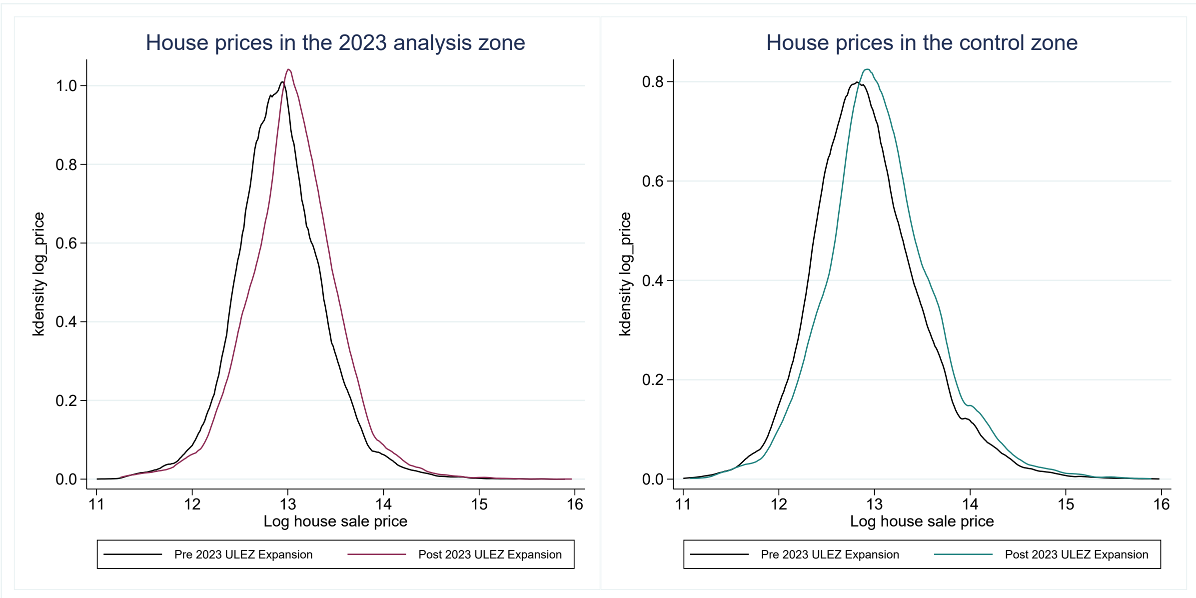
A more comprehensive approach involves estimating a dynamic version of this staggered DiD, alternatively known as an event study. I estimate the following dynamic TWFE specification via OLS:

$$y_{ipt} = \alpha_p + \delta_t + \sum_{j \neq j_{base}} \beta_j \mathbb{1}(t - G_p = j) + \rho \mathbf{X}_{it} + \varepsilon_{ipt}$$

where in addition to the above, G_p is the first quarter in which postcode sector p became part of the ULEZ, and j_{base} is a pre-period coefficient I normalise to zero (conventionally $j = -1$). Essentially, I replace the dummy $ULEZ_{pt}$ with a set of 'event-time' dummies that pick out all houses observed exactly j periods after they first came under the ULEZ. My coefficients of interest are the β_j, which isolate the average treatment effect on the treated (ATT) of being in the ULEZ on house prices j quarters after ULEZ was first implemented in the region, under a set of assumptions detailed below. This specification allows insight into the extensive-margin



(a)



(b)

Figure 3: (a) Distribution of log house sale price in the 2021 analysis and control zones, before and after the 2021 expansion; (b) As in (a), but replacing 2021 with 2023

effect, but also information on the 'intensive margin' about the dynamics of the effect over time. These dynamics are important from a policy perspective, enabling a distinction between short-run and long-run effects. In all regressions, standard errors are clustered at the postcode-sector level.

Certain assumptions must be made to ensure my identification strategy isolates causality. Most importantly, I assume that without treatment, the expected log house price would have evolved in the same way in the treated and control groups in the post-treatment period, conditional on my covariates. This is the (conditional) parallel trends assumption. This assumption cannot be directly tested, but if the pre-treatment dynamic coefficients are consistently near-zero, then relative prices did not evolve differently between treatment and control groups before treatment, suggesting the same would hold in the post-treatment period. The classic approach is to formally test the joint insignificance of these 'pre-trends', and apply the above logic. However, recent literature advocates against this test: it is low powered, and conditioning analysis on good pre-treatment behaviour implicitly introduces bias when estimating post-treatment effects (Roth, 2022). I therefore use them as no more than suggestive evidence that parallel trends holds. New research from Rambachan and Roth (2023) shows it is possible to perform inference under a more relaxed version of the parallel trends assumption, evaluating the robustness of results to violations of parallel trends. I deal with the assumption formally in this way.

Another important assumption is that there is no pre-treatment anticipatory response. Anticipatory effects may be present here, since the expansions were announced well before their implementation. However, if parallel trends holds, this assumption can be dealt with simply by considering 'treatment' as starting at the point where these anticipatory effects begin.

I also assume any effect does not come from confounding events in the region. The two events that threaten this assumption most are the opening of London's Elizabeth line, and the construction of the UK's HS2 railway. However, the effects of these events are both very local, so postcode-sector fixed-effects will capture most price variation they may cause. Furthermore, the Elizabeth line opened in 2022, while HS2 construction has been gradually implemented since 2020, out of time with the ULEZ expansions. The COVID pandemic may have influenced London's relative house prices, but again these effects would have been observed well before the expansions, leaving me satisfied that any observed effect comes from the ULEZ. Finally, to complete my set of assumptions, I assume no switching between treatment groups: houses clearly can't move.

The baseline TWFE model above is a workhorse model in applied microeconomics, but recent research has demonstrated that OLS estimates of the ATT in this specification are inconsistent when the effect is heterogeneous between treatment groups and/or over time. The issue is that the estimates are just weighted averages of all possible simple 2x2 difference-in-differences estimates between all groups at each time period, including 'bad' ones which compare later treated groups to earlier ones (Goodman-Bacon, 2021). Figure 3 suggests I have heterogeneity across groups, so my estimates of the ATT will be inconsistent in this framework. I therefore also use two new estimators of the β_j in the above model, both robust to this inconsistency problem: those from Abraham and Sun (2021), and Gardner (2021). Intuitively, Abraham and Sun solve this problem by estimating each 2x2 comparison individually and 'manually' putting them together correctly, while Gardner divides the process into two stages, estimating group and period effects using only untreated observations to avoid bad comparisons. Both will provide more reliable estimates than the basic TWFE model.

To deal with the issue of spillovers more flexibly, I also use the estimator from Butts (2024), which extends Gardner (2021). In this framework, I set a distance by which I assume spillovers end, and then estimate the fixed-effects only on observations beyond this distance from the ULEZ at the given time, to remove the influence of spillovers. I then don't have to drop the observations close to inner borders outlined in section IV: the issues they pose are dealt with in the estimation. This feature makes my estimated treatment effects more precise, and more representative of the effect across London. Furthermore, in the second stage, I can estimate not only the direct treatment effect, but also the effect of exposure to spillovers, allowing me to ascertain the effect the ULEZ had on house prices nearby but outside the zone.

Another important advantage of retaining these troublesome observations is that I am less restricted in where I assume spillovers end. Previously, assuming they ended at any further than 5km leaves me with few remaining observations in the zones after cutting out rings of this width (see Figure 2). I therefore extend this distance to 10km, reducing the spillover bias in my estimation of both the treatment effect and the spillover effect. I must then enlarge the outer control zone, as otherwise after the second expansion all houses are exposed, leaving none on which to estimate the later time fixed-effects. I therefore extend the outer zone to 15km outside the 2023 border, and again check robustness to failures of parallel trends using the methodology from Rambachan and Roth (2023). I still omit all houses within 5km of the 2019 ULEZ border, because of their fundamentally different treatment.

6 Results

Table 3 presents results from the baseline static staggered DiD, which shows a negative effect of the ULEZ on house prices. Without controls, in specification (1), prices are estimated to fall by 2.2%, but the more reliable specification with controls extends this estimate to 3.3%. Comparing the 2021 and 2023 expansions individually with the control zone through simple static DiD designs, in specifications (3) and (4) respectively, shows a much larger effect in the 2021 zone than the 2023 zone, aligning with the exploratory analysis. This is surprising given that more suburban regions further from central London are likely more car-reliant, but regardless, the effect is significantly negative in both areas. Heterogeneity analysis, in Appendix 2, indicates the negative house-price effect is concentrated within lower-priced houses, and particularly flats, which fits with the regressive nature of the charge. This Appendix also shows the results are robust to changes in the assumed distance thresholds by which the parallel trends assumption and any spillover effects end.

Figure 4 now plots the coefficient estimates from each of the three event-study estimators outlined above: standard TWFE estimated via OLS, and the robust estimators from Abraham and Sun (2021) and Gardner (2021). Appendix 3 gives a regression table containing these results, alongside all subsequent event-studies. I have addressed pre-emptive behaviour by normalising to zero the coefficient two quarters before first treatment (i.e. event time $j = -2$), since the coefficient estimates are remarkably flat up to event time $j = -1$, where they then exhibit a sudden drop. The figure has three important features. Firstly, in each case, the pre-treatment coefficients are generally close to zero and insignificant, suggesting no differential pre-treatment evolution of house prices and therefore supporting the parallel trends assumption. Second, the post-treatment coefficients are negative and generally significant, lying fairly consistently around -0.035 in the first two years after treatment, implying ULEZ led house prices in treated regions to fall by roughly 3-4% over that time. The charge, alongside any other negative effects of the ULEZ, have an overwhelmingly larger impact on housing valuations than the positive effects. Note that the absence of pre-trends supports the claim that the effects are unrelated to any COVID-induced suburbanisation: if they were, then since the UK's first lockdown occurred seven quarters before the first ULEZ expansion, a differential house-price evolution would be observed long before implementation, which is not found in the data. Finally, the treatment effect begins to recover over time when robustly estimated, and beyond two years after treatment they begin to become insignificant. I discuss this in section VII. Appendix 4 presents an alternative visualisation of these effects, plotting the evolution of region-wise control-adjusted means.

Two further points must be made. Firstly, although both expansions were announced long in

	(1)	(2)	(3)	(4)
ULEZ	-0.022***	-0.033***	-0.046***	-0.012*
	(0.0064)	(0.0059)	(0.0083)	(0.0064)
Treated Group	2021 & 2023	2021 & 2023	2021	2023
Postcode Sector FEs?	✓	✓	✓	✓
Quarter FEs?	✓	✓	✓	✓
Controls?		✓	✓	✓
Observations	481,579	481,579	238,039	357,795
R-squared	0.372	0.693	0.682	0.709

Robust standard errors, clustered at the postcode-sector level, in parentheses
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 3: Results from the static specifications

advance, the absence of a behavioural response until near implementation, as observed in Figure 4, is theoretically reasonable. Policies implemented by authorities frequently change in nature after their initial announcement - in fact, the 2023 expansion was challenged in the High Court, and only approved less than two months before it was implemented. Waiting until confident in the exact details of the expansion before adjusting behaviour seems rational in these circumstances. Notably, the sharp drop at event-time $j = -1$ is present not just in Figure 4, but also when comparing the 2021 and 2023 implementations individually with the control group – these dynamic DiD results are presented in Appendix 5. This consistency further supports the theoretical justification of no pre-emptive behaviour before $j = -1$.

Secondly, for event times $j > 5$, I only observe observations from the 2021 zone. This is a small issue, that I accept to avoid the alternatives: restricting my sample to only five periods after treatment obscures much of the dynamics of the effect, while presenting analysis for each zone separately sacrifices precision from not-yet treated zones serving as controls for already-treated zones. Regardless, even before $j = 5$, the treatment effect is significantly negative and large.

Figure 5 presents the results from the spillover-robust estimator from Butts (2024), which gives a less biased and more representative estimate of the overall ATT across Greater London. Encouragingly, these estimates align with what was observed before. The pre-treatment coefficient estimates for the ATT, as well as the spillover effect, are largely insignificant and non-trending, supporting parallel trends in both cases. The ATT is also similar in magnitude and dynamics to the event studies above. Once the full negative effect of the ULEZ is realised, the coefficient

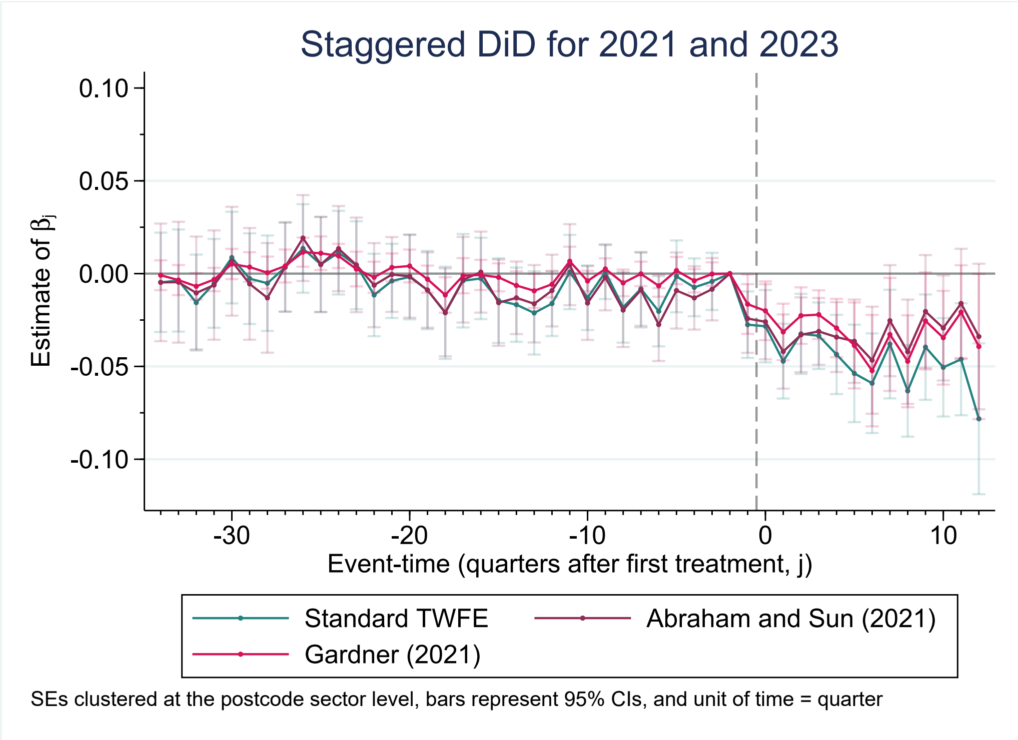


Figure 4: Estimates of β_j for each of the three event-study estimators above

estimates suggest it removes roughly 4% off the price of houses in the treated regions, and this effect remains fairly stable for about two years. Prices then begin to recover, as was also seen before.

The spillover effect is also significant in a few post-treatment periods, as well as visually convincing. The implementation of ULEZ leads house prices to fall by roughly 2% in areas within 10km of the border, and this effect appears to be sustained over a period of almost two years. It evolves in the same way as the direct effect does, suggesting the factors driving them are the same. Notably, I have fewer post-treatment coefficients for my spillover effects than for my direct treatment effects. To be exposed for longer than six quarters while still not treated, a house must be within 10km of the 2021 expansion zone but outside the 2023 expansion zone, because the 2023 expansion occurred six quarters after the 2021 expansion. Unsurprisingly, given Figure 1, few houses satisfy this criteria, so I observe a large drop in exposed houses for event-times $j > 6$. The estimates for these periods are therefore very imprecise and I omit them.

In sum, assuming parallel trends, there is substantial evidence to suggest that the ULEZ has a large negative impact on house prices, both within the zone and just outside it. However, while the parallel trends assumption is imperative in allowing a causal interpretation of the

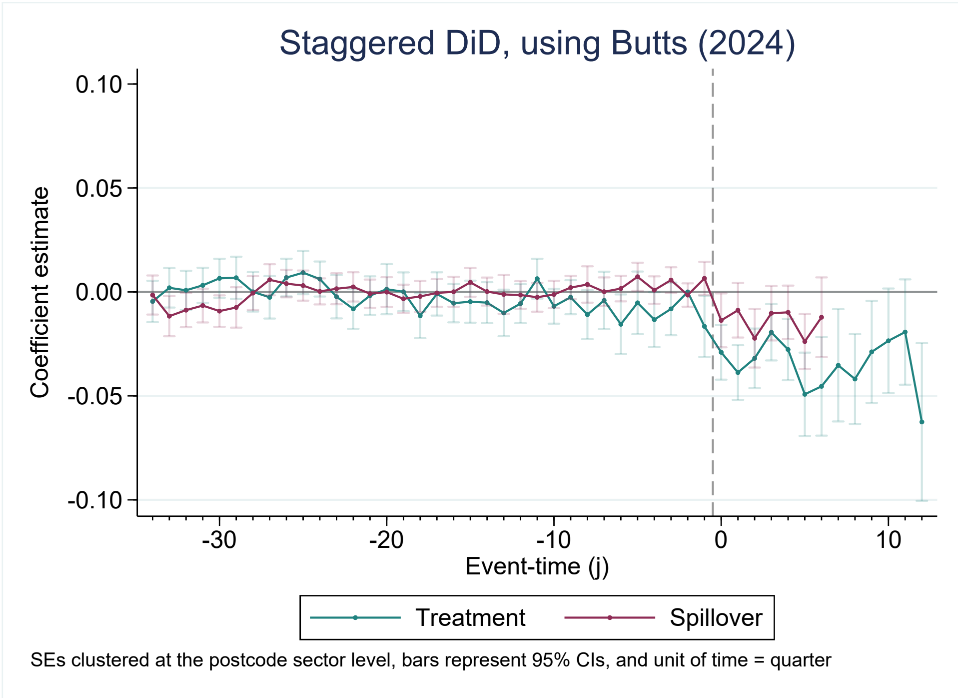


Figure 5: Estimates of the treatment and spillover effect, using Butts (2024)

observed effect, it is fundamentally untestable, as it is an assumption on the counterfactual. The classic approach to this issue is to test the joint insignificance of the pre-trends, and argue their insignificance implies these non-differential trends would continue past treatment, but as justified in section V, this approach has severe issues. I therefore do not present formal tests for the joint insignificance of my pre-treatment coefficients.

However, my results hinge heavily on parallel trends, so it is important to address their robustness to this assumption in some way. To do this, I use the methodology in Rambachan and Roth (2023). Intuitively, rather than assuming parallel trends, I relax the assumption to simply impose bounds on how badly parallel trends can fail in the post-period. With some regularisation steps as suggested by the paper, I find that if I were to allow deviations in parallel trends to occur repeatedly in the post-treatment period, each as large as the largest difference between consecutive coefficients in the pre-period, my estimates from Abraham and Sun (2021), Gardner (2021) and Butts (2024) all remain significantly negative at the 5% level. As such, while I could argue in the conventional way that parallel trends holds in my analysis, I have shown that the significance of my results are robust to large violations of this crucial assumption anyway. Full details, including an outline of the paper, a justification of why this alternative assumption is chosen, and a detailed explanation of my results under this weaker assumption, are presented in Appendix 6.

7 Discussion

The above analysis found that the ULEZ reduced house prices by roughly 3-4%, consistently among different estimators. As well as statistically significant, this effect is economically significant, with huge implications. Critically, it will impose huge welfare costs on London's population. The average price of houses sold in the expansion zones since the start of 2021 was around £690,000 - a 3.5% fall in this price corresponds to a roughly £25,000 windfall loss to the average homeowner in the region, over the space of a year. The median full-time salary in London in 2024 was £47,455 (ONS, 2024). The ULEZ therefore costed London's average homeowner roughly half the median salary in the region – a huge cost that cannot be ignored in favour of positive impacts of the ULEZ elsewhere. These welfare implications also raise concerns over inequality, particularly given that the effect falls disproportionately on lower-priced houses (see Appendix 2).

However, to understand the long-term impact of the ULEZ, the dynamics of the effect must be accounted for – in particular, the ATT begins to recover over time. This may be a product of asymmetric information transmission. Recalling the hedonic house valuation process, the driving restriction itself will be quickly incorporated into prices, because it is very quick to be observed and understood. Conversely, the positive effects the ULEZ has on air pollution and congestion are less obvious, and take longer to materialise, so will take longer to be reflected in house prices, which may explain the late signs of recovery. The long-run effect depends on the relative size of the positive and negative effects of the ULEZ, not their relative speed of emergence – the observed effect may just be temporary. Long-term conclusions cannot be made yet. See Appendix 7 for more detail.

My results are also highly applicable elsewhere. The house price effects of the ULEZ stem from the costly behaviour-altering effects it has on drivers of non-compliant cars. As such, the cities where emissions charges will have the worst effects are those most reliant on cars, so the external validity of my results can be ascertained by examining how London's car usage compares to other major cities. Figure 6 plots data collected from Eurostat, illustrating the distribution across European cities of the number of registered cars per 1000 inhabitants, and the share of journeys to work made by car. For each, I take the year in which data for London was last recorded. The results show that London is clearly one of the least car-dependent cities in Europe. This suggests that a ULEZ-type policy elsewhere in Europe would have even worse house-price effects than observed in London.

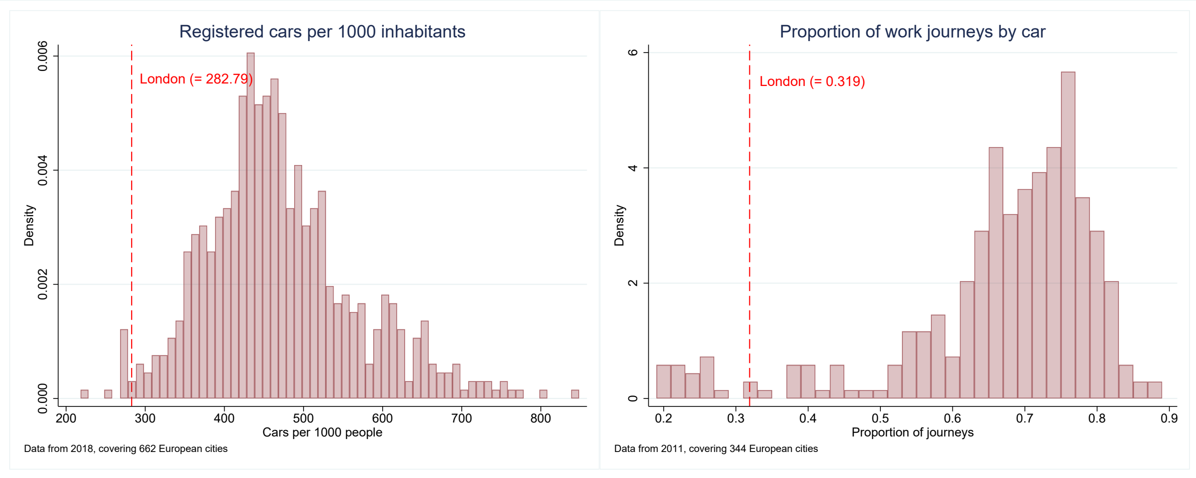


Figure 6: Histograms of transportation statistics for different European cities

8 Conclusion

This paper finds that the expansions of London's Ultra Low Emissions Zone in 2021 and 2023 caused house prices in the expansion regions to fall by roughly 3-4%. This corresponds to a windfall loss for the average London homeowner of about £25,000, or half the median local salary. This effect is similar among different estimators, and robust to more explicit ways of dealing with spillovers, as well as to a large weakening of the key identifying assumption of parallel trends. The ULEZ also had significant spillover effects, reducing house prices in regions within 10km of the zone by roughly 2%. London's relatively low car-dependence means these adverse short-term effects, even if rationalizable as simply a product of asymmetric information transmission, are externally valid. By highlighting the huge negative short-term side-effects of emissions-charges, this paper underscores the need for policymakers to consider complimentary measures to help residents deal with these effects.

My analysis could be improved in various ways. Firstly, more covariates would improve the precision of my estimates. More fundamentally, even while rigorously dealing with spillovers, I still implicitly assume they end at 10km, which is likely still too restrictive. This is a result of using London's suburbs as my control: I must cut spillovers off to ensure parallel trends holds. Alternatively, I could use regions in other large UK cities as my control, as in Gruhl et al (2022). Despite possibly jeopardising parallel trends, this constitutes a useful robustness check, which I will explore in future work.

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